

Coordination vs. Competition: Local Government Centralization and Metropolitan Growth

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Abstract

How does the organization of local government affect population growth? Is growth promoted in centralized governments concentrated in a few large units or decentralized governments consisting of many smaller units? The literature provides conflicting answers to these questions. On one hand, decentralization can promote competition between local governments, providing a market-like mechanism to obtain efficiency in the supply of public goods. On the other, externalities can foster coordination failures that undermine competition. Which of these two opposing forces is more important for local government? In this paper, I use local government employment data from the metropolitan statistical areas of the United States and an IV strategy to estimate the causal effect of local government centralization on growth. I then examine the effect of centralization on local government employment, finances, and the siting decisions of data centers to determine which mechanisms drive the effect on growth. My instrument is derived from a complete history of post offices in the United States and leverages a historical policy change, the implementation of Rural Free Delivery in 1902, to measure the presence of small communities and dispersed populations. I show that the instrument shifts the organization of local government toward decentralization and argue that this shift is plausibly exogenous to the modern outcomes I examine. Using the variation induced by the instrument, I find that centralization causes higher population growth rates, higher total local government employment, more tax revenue to be collected, and fewer data centers to locate in the metropolitan area. Based on these empirical results, I argue that centralized local governments best promote growth, with the most likely mechanism being the resolution of coordination failures.

1 Introduction

A longstanding question in economics asks how government should be organized. Because public goods lack market-based provision mechanisms (Musgrave (1939); Samuelson (1954)), economists have long worried about the efficiency of their supply. Foundational contributions by Tiebout (1956) and Oates (1972) argue that government organization itself shapes these outcomes, framing the central debate: should public services be centralized into a few large jurisdictions, or decentralized across many smaller ones?

This question has direct policy relevance, not just academic interest. Many countries have actively restructured local governments in recent decades. Between 2008 and 2017, fifteen European countries implemented large-scale municipal reforms that increased centralization through mergers of government units (Swianiewicz (2018)). Japan undertook a similar reform in the early 2000s, reducing its number of municipalities from 3,232 to 1,719 through mergers (Miyazaki (2014)). In the United States, voters have repeatedly considered local government consolidation, including 166 city-county merger referenda since 1815 (Hall, Matti, and Zhou (2020)), alongside numerous smaller merger and annexation decisions. These merger events represent discrete jumps in centralization. Therefore, understanding the economic consequences of centralization is essential for evaluating why such reforms occur and how they shape outcomes in metropolitan areas.

Rather than studying individual merger events, in this paper I exploit variation in centralization across the 379 metropolitan statistical areas (MSA) in the continental United States. I measure centralization three different ways using data from the Annual Survey of Public Employment and Payroll, which reports employment for every local government unit in the United States. First, I measure the percentage of total MSA employment in the largest local government unit, with a higher share indicating higher centralization. Second, I measure centralization with the number of local governments within an MSA, with more governments leading toward higher decentralization. The third way I measure centralization is through a Herfindahl–Hirschman index I construct based on governments’ shares of total MSA-level public employment. I find that regardless of the measure used, the qualitative effect of centralization remains consistent.

The degree of centralization I measure in each MSA captures a tradeoff between centralization and decentralization. An MSA’s implied position in the distribution of centralization across MSAs is then not randomly assigned, and may be influenced by many observed and unobserved factors. My measures of centralization are therefore likely endogenous, and neglecting this endogeneity may lead to biased estimates of the effect of centralization. To overcome this problem and identify the causal effect of centralization, I use an instrumental variables strategy based on the introduction of Rural Free Delivery in 1902. Using a complete census of U.S. post offices from Blevins and Helbock (2021), I measure post office closures between 1902 and 1910 induced by the implementation of the new legislation, which mechanically reduced post office revenues. The shock induced by Rural Free Delivery disproportionately affected post offices serving small and spatially dispersed communities, making them a proxy for settlement dispersion and the presence of small population centers. These features strongly predict modern local government decentralization but likely affect my out-

comes only through their influence on jurisdictional structure, not through contemporaneous economic channels.

I first show that greater local government centralization causes higher MSA population growth. The result holds through multiple time horizons with the shortest spanning 2000 to 2020, the longest spanning 1950 to 2020, and for all decennial periods in between. Classical political economy claims that “the most decisive mark of the prosperity of any country is the increase of the number of its inhabitants” (Smith (1776), Book I, Chapter VIII), motivating me to treat population growth as a central outcome of interest that all other results seek to explain. Modern spatial equilibrium models formalize this intuition, with population growth reflecting increased expected utility for residents (Rosen (1979); Roback (1982); Moretti (2011); Redding and Rossi-Hansberg (2017)). The positive causal effect of centralization on population growth therefore suggests that, on average, centralized local governments offer higher expected utility to residents. This finding contrasts with canonical decentralization-based theories of government organization (Tiebout (1956); Ostrom, Tiebout, and Warren (1961); Brennan and Buchanan (1980)) which emphasize competition between local governments as a disciplining force and predict that greater decentralization, not centralization, would lead to higher population growth.

Why do centralized MSAs offer higher expected utility and therefore have higher population growth? I interpret the link between local government organization and expected utility as stemming from differences in the efficiency with which public goods are supplied, with greater efficiency leading to higher expected utility. Through what mechanisms are centralized MSAs better able to provide public goods? An answer to this question can potentially be found in the models that promote decentralization. The formulations of these models generally rule out both economies of scale and coordination failures stemming from interjurisdictional externalities. Their omission suggests that they may be two primary channels through which centralization may instead generate gains. In the remaining empirical exercises, I test whether these two mechanisms explain the higher growth I find in more centralized MSAs.

I confront two primary difficulties in testing for economies of scale: first, I do not observe public good output directly; second, public goods are bundles of goods and services. Even if all dimensions of the public good bundle were observable, I would still confront the issue that the observed allocations between different dimensions reflect tradeoffs made between them. Higher quality roads found in one MSA may reflect residents’ stronger preferences for roads over parks, not the efficiency of their supply. To partially overcome this problem, I focus on local government employment and find that more centralized MSAs use more labor inputs. This pattern contradicts a technical efficiency mechanism in which centralization primarily reduces labor inputs and increases expected utility by passing cost savings on to residents. If technical efficiencies stemming from economies of scale were the dominant channel, centralization would reduce, not increase, labor inputs.

Second, I find that more centralized local governments collect more tax revenue conditional on population. This result is consistent with the predictions of tax competition models, which constitute an important case of coordination failures. By splitting taxing authority across jurisdictions, decentralized MSAs intensify competition for a mobile tax base. In common models, competition between local governments fails due to a fiscal externality:

individual jurisdictions do not internalize that increases in their own tax rates primarily reallocate the tax base rather than eliminate it. This failure to coordinate tends to suppress equilibrium tax rates (Wilson (1986); Zodrow and Mieszkowski (1986); Wildasin (1988)), and is consistent with the empirical finding that more centralized MSAs are able to increase their tax revenues.

I next use the exogenous variation in centralization from the IV strategy to test whether either of the plausible mechanisms has an independent effect on population growth. If economies of scale, as measured by local government employment, or coordination failures, as measured by tax revenues, were a dominant mechanism, they should account for at least part of the population growth associated with centralization.

When examining economies of scale, I find no independent role for local government employment on growth. The estimated effect of centralization on growth remains unchanged when employment is included, suggesting that if the gains from centralization operate through economies of scale, they must do so in a way that is not reflected in labor inputs. In contrast, I estimate that tax revenue has an independent effect on growth, and that its inclusion in the growth specification reduces the role of centralization. This suggests that I cannot rule out the causal channel of centralization increasing fiscal capacity, and fiscal capacity accounting for population growth. As I do not have a separate instrument for either local government employment or tax revenue, these results are not causal and are only suggestive of which mechanism is dominant.

In a final empirical exercise, I examine location decisions made by firms constructing data centers. Data centers are relatively new, highly mobile, capital-intensive, and particularly sensitive to local tax policy, making them an ideal test case for the presence of coordination failures among local governments. If coordination failures result in lower equilibrium tax rates, firms constructing data centers have an incentive to disproportionately locate in decentralized MSAs to take advantage of the lower effective tax burdens. Consistent with this mechanism, I find that decentralized MSAs are more likely to host at least one data center, suggesting that coordination failures are pronounced enough to affect firm decision-making.

Taken together, the results most strongly suggest that the gains in population growth found in centralized MSAs are caused by the alleviation of coordination failures. Because interactions among local governments are inherently spatial, externalities are particularly salient within MSAs. Further, as emphasized by Sinn (1997), governments operate under the selection principle; they disproportionately provide goods and services whose provision is marked by externalities, as their existence undermines market-based provision. In short, the results I find suggest that, by internalizing externalities, centralized local governments mitigate coordination failures and increase the efficiency of public good provision, thereby allowing for higher expected utility for residents, which leads to higher population growth.

This work contributes to several literatures, beginning with research on political mergers. Because political mergers generate discrete increases in local government centralization, estimating the causal effect of centralization is central to evaluating these reforms. Existing studies, largely based on case studies and quasi-experimental designs, examine a wide range of institutional contexts, including Israel (Reingewertz (2012)), China (Tang and Hewings (2017)), Denmark (Blom-Hansen et al. (2016)), Japan (Miyazaki (2018)), Germany (Egger

et al. (2018)), and Iceland (Karlsson and Eythórsson (2022)). Recent syntheses of the literature emphasize the highly heterogeneous and often contradictory results across settings and outcomes (Zheng, Sun, and De Witte (2023)). In contrast, I estimate the causal effect of centralization across relatively homogeneous U.S. metropolitan areas using variation generated by a historical policy shock. By abstracting from discrete merger events, this approach mitigates concerns about institutional heterogeneity and endogenous selection that complicate merger-based analyses.

This work also relates to the literature on government centralization and institutional outcomes. A large macro-level literature studies how cross-country variation in fiscal and political centralization between national and sub-national governments affects growth and institutional performance (Barro (1991); Weingast (1995); Sala-i-Martin (1997); Davoodi and Zou (1998); Treisman (2000); Rodden (2004); Iimi (2005); Baskaran and Feld (2013)). In contrast to my findings, this literature often concludes that decentralization promotes economic growth (Dillinger (1994)). One interpretation is that the relevant tradeoffs differ by scale. At the national level, decentralization may intensify competition across large, geographically separated jurisdictions (e.g. states) where spillovers are less prominent. In contrast, at the metropolitan level the dominant concern is internalizing spillovers among closely linked governments, making centralization more effective.

A second strand examines similar questions using within-country variation in governmental structure and centralization. Examples include Hoxby (2000) on educational performance; Stansel (2005) on population growth; Hatfield and Kosec (2013) (2019) on wage growth and pollution; Hatfield, Kosec, and Rodgers (2024) on housing prices; Hobbs, Millsap, and Stansel (2019) on economic development; and Bianchi, Giorcelli, and Martino (2023) on labor market outcomes. This paper fits most closely within this strand of the literature. However, existing studies rely on a wide range of centralization measures and sources of identifying variation that do not map cleanly to changes in local government centralization as measured here. Differences in how centralization is defined and identified may therefore account for the divergence between prior findings and the results in this paper.

The remainder of the paper proceeds as follows. Section 2 presents the data on local government organization, including the measurement of centralization, the outcomes I study, and the construction of controls capturing systematic heterogeneity across metropolitan areas. Section 3 outlines the empirical strategy and describes the instrumental variables used for identification. Section 4 reports the causal estimates and discusses their implications for understanding the mechanisms of centralization. Section 5 concludes.

2 Data

2.1 Local Government Data and Measurement

The primary unit of analysis in this study is the metropolitan statistical area (MSA), as defined by the Office of Management and Budget (OMB). An MSA consists of a group of counties organized around an urban core with substantial economic integration, measured

primarily through commuting flows. The median MSA contains two counties, and 149 of the 379 MSAs consist of a single county; the Atlanta–Sandy Springs–Roswell MSA contains the most counties with 29. Because commuting patterns and urbanization evolve over time, MSA boundaries change across decades. To maintain a consistent geographic unit, I fix MSA boundaries at their 2020 definitions and use these boundaries to harmonize all historical variables. This approach allows changes over time to be interpreted as occurring within the same geographic areas.

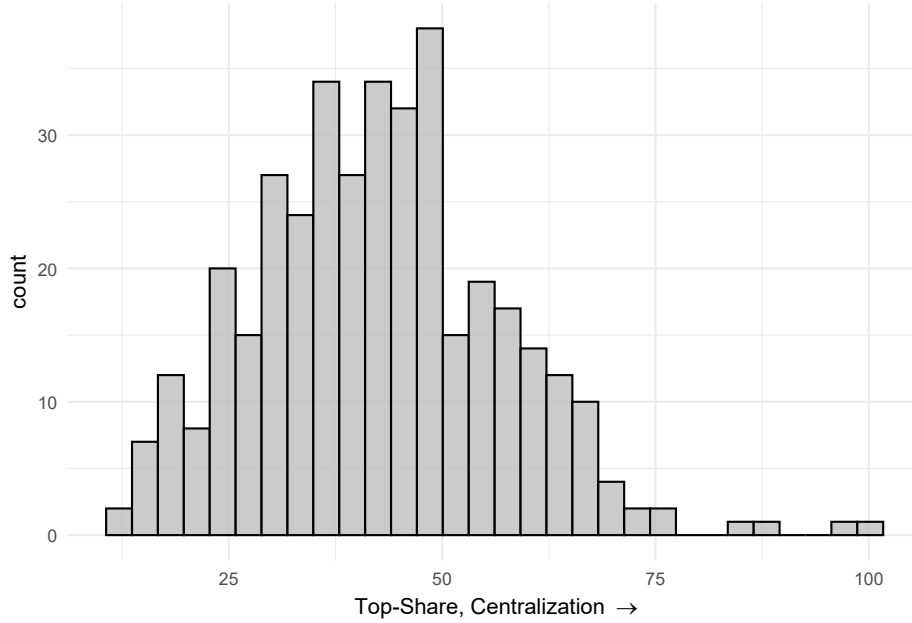
I measure local government organization within MSAs using data from the 2007 Annual Survey of Public Employment and Payroll (ASPEP), the earliest wave without broad aggregation at the individual government-unit level. ASPEP conducts a full census of local governments only every five years, and earlier waves such as 1997 and 2002 suffer from substantial under-reporting and aggregation across government units, making them unsuitable for measuring local government structure at the MSA level.

ASPEP reports employment for each local government unit. I drop all government units that report no employment. The remaining local governments are classified into five types: counties, municipalities, townships, school districts, and special districts. ASPEP also reports employment across 32 functional categories, such as policing or roadworks. I make two main exclusion decisions. First, I drop school districts from the analysis, as education operates under distinct governance, financing, and labor-market regimes and represents a separate institutional margin from general-purpose local government organization. To harmonize across MSAs, I also drop all employment in education-related functional categories. Second, I drop all local government employment in hospitals. Aside from these adjustments, I abstract from functional allocation of employment, with one exception. I construct an indicator variable for whether an MSA has any local government employment in ports, as port employment captures geographic heterogeneity related to coastal and river access.

2.2 Local Government Centralization

In this study, the main variation I examine is the degree of centralization within an MSA, which I measure in three ways. The first way I measure centralization is through the share of total MSA local government employment in the largest government unit, which I call *Top-Share*. I normalize the Top-Share measure to lie between 0 and 100. As the measure approaches 100, the majority of an MSA’s local government employment is concentrated into a single government unit, indicating a high degree of centralization. Conversely, as the Top-Share measure approaches 0, employment is spread across many different units, indicating a high degree of decentralization. Figure 2.1 displays a histogram of the Top-Share measure across MSAs.

Figure 2.1: Distribution of Top Share Measure of Centralization



Notes: Histogram of the Top-Share measure of local government centralization across U.S. metropolitan statistical areas (MSAs). Top-Share is defined as the percentage of total local government employment in the single largest government unit within the MSA. The horizontal axis reports the Top-Share value and the vertical axis reports the number of MSAs in each bin.

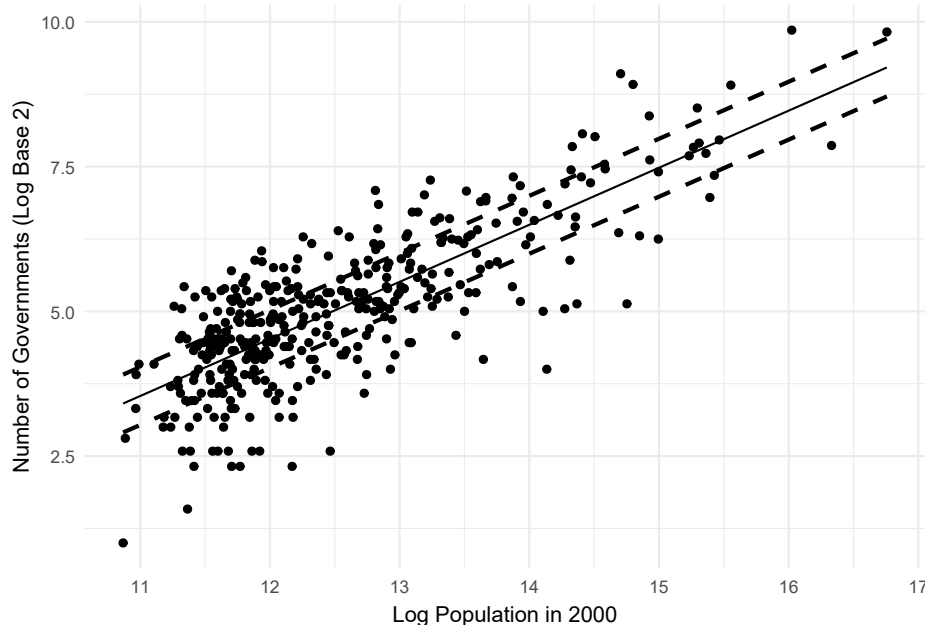
The figure shows a large degree of variation. The median MSA has a Top-Share of 42.7%, and the interquartile range is 20.3%. Another dimension of heterogeneity in the Top-Share measure is based on what type of government (city, township, county, or special district) is the largest in the MSA. In 54% of MSAs, the largest government entity by employment is a county, and in the remaining 46% it is a city. The means between these two groups are 44.2% and 40.6% respectively, and while statistically different from each other, do not indicate a large source of variation.

The Top-Share measure of centralization is especially convenient for considering counterfactual political mergers. For example, the government of St. Louis City is the largest local government in the St. Louis MSA, accounting for 19% of employment. Throughout the region's history, there have been several unsuccessful attempts to merge the government of the city with the second largest local government, St. Louis County. A successful merger between the two governments would increase centralization, as measured by Top-Share, in the MSA by 11 percentage points. The counterfactual effect of this merger on an outcome can be approximated by extrapolating the marginal effect, a one percentage point increase, estimated with the Top-Share measure.

The next way I measure centralization is through the *Number of Governments*, which measures the number of government units with at least one full-time employee in the MSA. Figure 2.2 plots this measure against log population in 2000. The vertical axis reports the number of local governments measured in log base 2, while the solid line shows the fitted

OLS relationship between population and the number of governments. The dashed lines indicate a one-half unit deviation from the fitted line.

Figure 2.2: Number of Governments by Population



Notes: Scatter plot of the number of local governments in each MSA against log population in 2000. The vertical axis reports the log base-2 number of local governments with at least one employee, and the horizontal axis reports log population. Each point represents an MSA. The solid line shows the fitted OLS regression line and the dashed lines show the fitted line shifted by ± 0.5 units.

Figure 2.2 displays a strong positive relationship between the number of governments in an MSA and population. The normalization to log base 2 is both conceptually appealing and overcomes the unrealistic restriction of constant marginal effects for each additional government imposed when measuring in levels. A one unit shift, visualized as the vertical distance between the dashed lines in Figure 2.2, is equivalent to doubling the number of governments in an MSA, capturing a large shift towards decentralization while remaining well within sample.

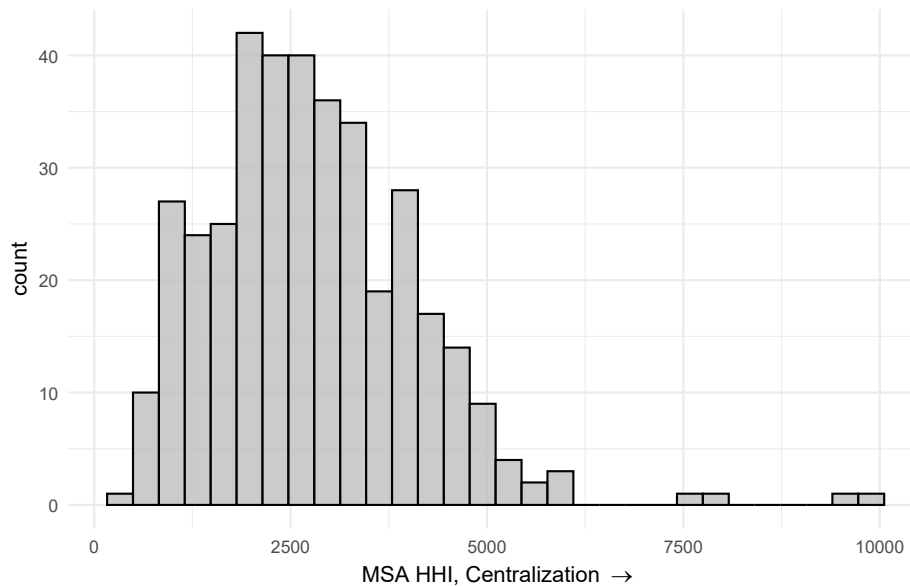
The number of governments measure emphasizes that centralization, across all measures, should be viewed conditional on population. A fixed number of governments may be classified as decentralized in a MSA with a low population, but centralized in a MSA with a larger population. This idea can be seen graphically in Figure 2.2, where comparisons in centralization between MSAs should be made vertically instead of horizontally

The third and final way I measure centralization is through a Herfindahl–Hirschman Index (HHI) calculated from local government employment shares. I compute the *MSA HHI* as:

$$HHI_m = 10,000 \cdot \sum_{i=1}^{N_m} \left(\frac{\text{Employment}_{im}}{\sum_k^{N_m} \text{Employment}_{km}} \right)^2$$

where Employment_{im} denotes employment in government unit i within MSA m , and N_m is the number of local government units in the MSA, equivalent to the Number of Governments measure. The index approaches 10,000 when local government employment concentrates in a small number of large governments, indicating centralization. As MSA HHI approaches zero, employment disperses across many small units, indicating decentralization. Figure 2.3 displays the distribution of MSA HHI across MSAs in a histogram.

Figure 2.3: Distribution of MSA HHI Measure of Centralization



Notes: Histogram of the MSA Herfindahl-Hirschman Index (HHI) constructed from local government employment shares. The horizontal axis reports the HHI value (sum of squared employment shares multiplied by 10,000) and the vertical axis reports the number of MSAs in each bin.

2.3 Outcomes

2.3.1 MSA Level Population Growth

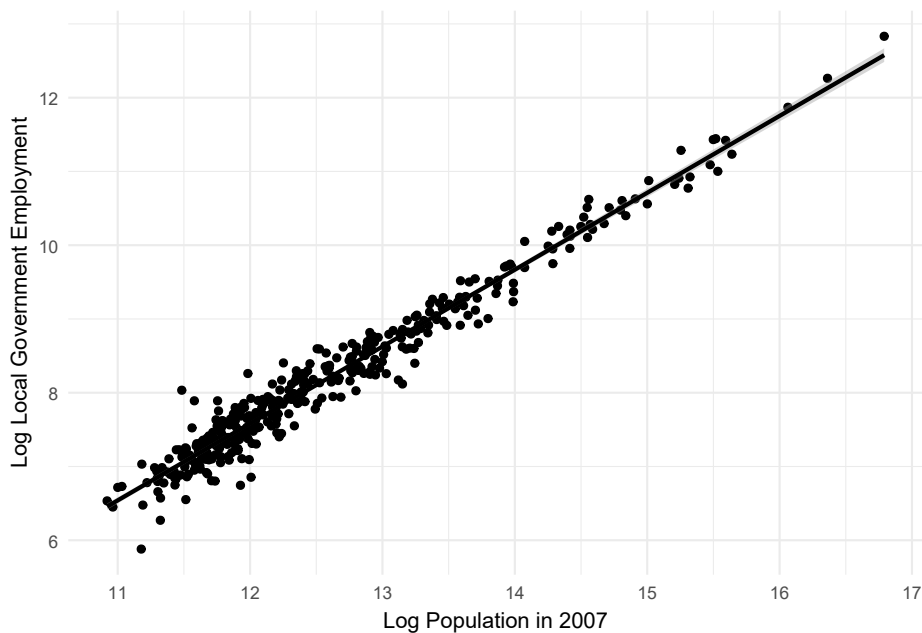
I measure population growth at the MSA level using log growth rates, defined as the log of population in the post period divided by population in the base period. I always consider the post period to be 2020. To measure the population in each MSA, I retrieve population statistics for all US counties from the decennial Census for 2020, 2000, 1990, 1980, 1970, 1960, and 1950. I then aggregate county populations using the 2020 OMB MSA definitions for both time periods. From this data I create six population growth statistics, one for each year other than the common post period of 2020. Between 2000 and 2020, the mean log growth rate across MSAs is 0.17 corresponding to roughly 18% population growth, with a standard deviation equal to .15, indicating substantial heterogeneity across MSAs. The growth rates derived using earlier base years are almost always higher, accounted for by

the sustained population growth in MSAs throughout time. In some later applications, I annualized growth rates by dividing by the number of years between 2020 and the base year.

2.3.2 Local Government Employment and Finances

I measure local government employment at the MSA level using the same data that I use to create the centralization measures. Figure 2.4 plots log local government employment by log population in 2007. The figure displays a strong log-linear relationship, motivating a log-log empirical strategy detailed in Section 3. As MSAs vary widely in size, changes in local government employment are likely scale dependent and therefore should be measured proportionally. For example, an exogenous shock to local government employment may have relatively homogeneous proportional effects, but the level may be wildly different depending on the existing number of local government employees. Similarly, a given change in centralization is unlikely to generate the same absolute effect on employment in a small MSA as in a large one, but it may generate a comparable percentage change. The log-log specification therefore aligns naturally with both the scale of the data and the mechanisms through which centralization likely operates.

Figure 2.4: Local Government Employment by Population



Notes: Scatter plot of local government employment against population across MSAs. The horizontal axis reports log population in 2007 and the vertical axis reports the log number of local government employees. Each point represents an MSA. The solid line shows the fitted linear relationship between the two variables.

Along with the employment data, I measure total local government tax revenue using the 2007 Census of Governments (CoG). The CoG is similar in structure to the ASPEP. Both report data for each local government unit in the United States and the timing for full

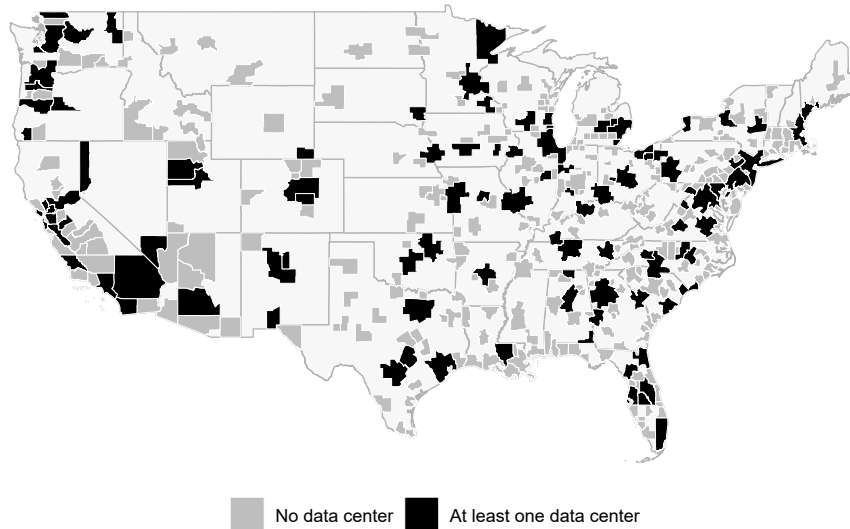
censuses of each are the same. While ASPEP focuses on employment, CoG focuses on finance. Like the employment data, I measure MSA total tax revenue as the sum of tax revenues from all local governments in the MSA.

2.3.3 Data Centers

The location of data center facilities is the final outcome that I investigate. I use data from the IM3 Open Source Data Center Atlas (Mongird et al. (2025)) hosted by the Office of Scientific and Technical Information within the U.S. Department of Energy. The data reports 1242 unique data center facilities along with the county within which they are located. To capture this extensive margin, I create an indicator for each MSA that reports whether there exists at least one data center facility in the MSA. From the data, I find that 27% of all MSAs host at least one data center facility.

Figure 2.5 displays the geographic distribution of the extensive margin of data center facilities across the U.S. MSAs. MSAs shaded in black have at least one data center facility, while those shaded in grey have no data center facility presence.

Figure 2.5: Geographic Distribution of Data Centers Across U.S. MSAs



Notes: Map of U.S. metropolitan statistical areas indicating the presence of data center facilities. MSAs shaded in black contain at least one data center facility while MSAs shaded in gray contain none. Alaska, Hawaii, and U.S. territories are excluded.

2.4 Additional Data

In its most basic form, my empirical strategy detailed in Section 3 consists of regressing the outcomes discussed in Section 2.3 on the centralization measures discussed in Section 2.2.

In this section, I discuss additional data that I employ as controls in those regressions. For each of the three centralization measures, I report three different specifications, with the first baseline specification containing only population, land area, and the centralization measure.

The next specification includes a set of largely time-invariant differences between MSAs. These include the port indicator variable discussed in Section 2.1 and the type of government used in creating the Top-Share centralization measure. I also include the first year that a city was incorporated in the MSA, taken from Goodman (2023), and Census Division level fixed effects to capture geographic heterogeneity. The final measure I include in the time-invariant specifications is a measure of topographical heterogeneity called the Terrain Ruggedness Index as discussed in Nunn and Puga (2012) and similar to the measure used in Saiz (2010). This variable captures differences in elevation within each MSA, with higher values of the index indicating greater local changes in elevation.

The third specification includes a set of modern covariates capturing economic, demographic, and housing market conditions within MSAs measured in 2000. The modern controls include median household income, the poverty rate, the unemployment rate, and the labor force participation rate. Together, these variables capture differences in local economic conditions and labor market strength that may correlate with both centralization and the outcomes of interest. I include median house prices to account for variation in housing market conditions that may independently influence population movements and local fiscal capacity. I also include demographic controls capturing the median age, the percent of the population that is not white, the share of renters, the share of adults with a college degree, and the percentage of the workforce in the manufacturing sector and in the technology sector to capture the industrial composition of the MSA. Finally, I measure variance in population density by comparing block level population density to the average density implied by population and land area alone. Higher values in this measure indicate that the observed population density in 2000 differs more from the uniform density.

Table 2.1 reports summary statistics for these variables.

Table 2.1: Summary Statistics

Statistic	Mean	St. Dev.	Min	Max	Median
Local Government Employment [†]	9,405.72	26,339.38	358	374,052	2,807
Log Tax Revenue ^{††}	12.39	1.31	10.24	17.85	12.10
Data Center Facilities [‡]	0.27	0.45	0	1	0
Top-Share [‡]	42.62	14.34	11.83	99.85	42.73
Number of Governments [‡]	55.04	90.39	2	927	32
MSA HHI [‡]	2,778.57	1,335.18	409.13	9,970.97	2,670.44
Log Land Area [§]	8.40	0.86	5.93	11.16	8.39
Terrain Ruggedness Index ^{§§}	5.59	6.88	0.14	49.73	2.53
First Year of City Incorporation [¶]	5.03	0.27	3.87	5.91	5.04
Indicator: Ports Employment [‡]	0.21	0.41	0	1	0
Median House Price (1000's)*	104.41	43.14	42.80	418.35	94.12
Median Income (1000's)*	39.66	6.66	24.86	73.82	38.71
Median Rent*	537.86	105.25	361.00	1,172.12	516.22
Poverty Rate*	12.30	4.10	4.38	35.45	11.57
Percent Unemployed*	5.89	1.87	2.64	13.57	5.57
Labor Force Participation Rate*	48.66	4.72	29.75	59.19	49.28
Median Age*	35.32	3.88	23.37	54.30	35.51
Percent Non-White*	40.72	12.42	22.14	96.45	37.51
Percent Renters*	31.86	5.82	13.51	49.03	31.19
Percent College Education*	13.16	2.55	6.13	21.47	13.11
Percent Manufacturing Employment**	15.89	8.22	1.98	51.82	14.98
Percent Technology Employment**	8.25	3.76	1.93	30.20	7.38
Population Density Variance*	2.17	0.71	0.70	4.31	2.02

[‡] - IM3 Open Source Data Center Atlas

[¶] - Blevins (2021)

^{§§} - U.S. Geological Survey

[†] - US Census Bureau, Annual Survey of Public Employment and Payroll 2007

^{††} - US Census Bureau, Census of Governments 2007

* - US Census Bureau, 2000 Decennial Census

[§] - US Census Bureau, TIGER/Line Shapefiles

** - US Census Bureau, County Business Patterns

3 Estimation Strategy

3.1 Specifications

In this section I discuss the specifications used in my estimation strategy. The centralization measures, Top-Share, Number of Governments, and MSA HHI, are denoted by C_m , and their effect on the outcome of interest is γ . The time-invariant MSA characteristics are collected in W_m^T , and the modern controls are collected in W_m^C . Throughout, I assume that there is an unobserved determinant of the outcome ξ_m that may be correlated with the centralization measure C_m . Because C_m and ξ_m are correlated, the effect of centralization on the outcome, γ , cannot be estimated without bias through OLS. I overcome this difficulty with an IV strategy discussed below.

3.1.1 MSA Level Population Growth

I measure MSA level population growth as the log ratio of the MSA's population in two periods. The post period is always taken to be 2020. This specification comes from a multiplicative growth process given as:

$$P_{m,2020} = P_{m,t}^\beta \cdot L_m^\lambda \cdot \exp(\epsilon_{m,t})$$

Where $P_{m,t}$ measures the population in MSA m at time t , β is a parameter that governs how growth scales with starting population size, L_m and λ measure land area in the MSA and its effect on growth, and $\epsilon_{m,t}$ is a MSA specific shock. Taking the logarithm and moving one unit of the $P_{m,t}$ term to the left hand side yields:

$$\log \frac{P_{m,2020}}{P_{m,t}} = (\beta - 1) \log P_{m,t} + \gamma \log L + \epsilon_{m,t}$$

To create my empirical specification, I decompose the error term into its mean, the effect of centralization, the effect of the time-invariant MSA characteristics, the effect of the modern controls, the unobserved determinant of growth, and a mean zero idiosyncratic term yielding:

$$\epsilon_{m,t} = \alpha + \gamma C_m + \delta^T W_m^T + \delta^C W_m^C + \xi_m + u_{m,t}$$

Substituting in the error term decomposition, I arrive at the empirical specification that I take to the data given by:

$$\log \frac{P_{m,2020}}{P_{m,t}} = \alpha + (\beta - 1) \log P_{m,t} + \lambda \log L_m + \gamma C_m + \delta^T W_m^T + \delta^C W_m^C + \xi_m + u_{m,t}$$

To estimate the effect of centralization through different time periods, I change the base year captured by t . As growth has been generally positive throughout time, when comparing across different base years I annualized the effect by dividing by the elapsed time in years, $2020 - t$.

3.1.2 Local Government Employment and Finances

The specifications I use for local government employment and finances are nearly identical, and mirror the specification for population growth. Focusing on local government employment, Figure 2.4 displays a strong log-linear relationship with population. The pattern suggests that local government employment scales proportionally with population. I formalize the relationship between population and local government employment as:

$$\text{Employment}_m = \alpha \cdot P_{m,2007}^\beta \cdot L_m^\lambda \cdot \exp(\epsilon_m)$$

Where Employment_m captures total local government employment in MSA m , and α captures the average number of local government employees needed to serve a population of a

given size. From here, I take the logarithm and perform an identical error-term decomposition as in the growth equation, yielding:

$$\log \text{Employment}_m = \log \alpha + \beta \log P_{m,2007} + \lambda \log L_m + \gamma C_m + \delta^T W_m^T + \delta^C W_m^C + \xi_m + u_m$$

The estimating equation used for tax revenue is nearly identical. Log tax revenue plays the role of log employment, and median income is included in all specifications to handle local price differences across MSAs Moretti (2013).

3.1.3 Data Center Locations

I estimate the effect of centralization on the extensive margin of data center location. The extensive margin captures whether an MSA hosts at least one data center facility. Let $D_m = 1$ if MSA m has at least one data center and let $D_m = 0$ if not. I take the following equation to the data:

$$D_m = \alpha + \gamma C_m + \beta P_{m,2020} + \delta^T W_m^T + \delta_m^C W_m^C + \xi_m + u_m$$

3.2 Rural Free Delivery Instrument

In the previous specifications, OLS cannot identify the effect of centralization because the centralization measure is possibly correlated with an unobserved determinant of the outcome. If this correlation exists, the OLS estimate conflates the effect of centralization with the effect of the unobservable, and is therefore biased. To overcome this issue, I employ an IV strategy. The instrument is constructed from a complete history of U.S. post offices compiled by Blevins (2021). I measure the log number of post office closures between 1902 and 1910, a period immediately following the implementation of Rural Free Delivery (RFD).

In 1893, the 53rd United States Congress enacted legislation requiring post offices to provide mail delivery without charge. Prior to this reform, while mail collection at post offices was free, delivery to residences or businesses could be fee-based. With full implementation achieved in 1902, post offices were required to deliver mail, free of additional charge, to all mailboxes.

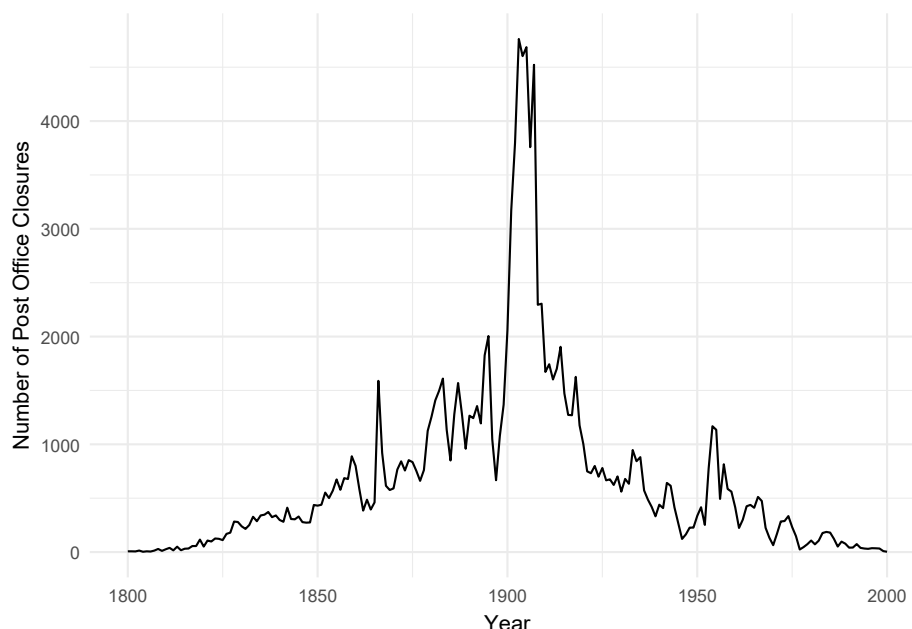
From the perspective of individual post offices, RFD generated a sharp reduction in revenue without a corresponding reduction in operating costs. As a result, post offices that were previously marginally viable were more likely to close. These closures disproportionately occurred among smaller post offices and those serving dispersed populations, so the number of post office closures within an MSA during this period reflects pre-existing patterns of population dispersion and the prevalence of small settlements.

These underlying conditions are closely linked to the subsequent organization of local governments. A small community that previously maintained a post office was more likely to incorporate later as its own city. A community far from the primary city of the region was less

able to rely on public services from the primary city and therefore more likely to incorporate in order to collectively provide these services, contributing to greater decentralization. In this sense, post office closures capture historical determinants of local government structure rather than contemporaneous economic or demographic conditions.

Figure 3.1 displays the number of post office closures in the United States between 1800 and 2000. The y-axis measures the number of post office closures, and the x-axis reports the year.

Figure 3.1: Post Office Closures: 1800-2000



Notes: Time series of annual U.S. post office closures from 1800 to 2000. The horizontal axis reports the calendar year and the vertical axis reports the number of post office closures occurring in that year. The figure highlights the spike in closures following the introduction of Rural Free Delivery in the early 1900s.

The figure shows a pronounced spike in closures during the period associated with the implementation of Rural Free Delivery, corresponding to the years used to construct the instrument.

3.3 First Stage and Robustness

I report the first-stage results of the 2SLS estimator in Table 3.1. Columns 1 through 3 report the effect of the instrument on the three centralization measures, with Column 1 reflecting the Top-Share measures, Column 2 reflecting the Number of Governments, and Column 3 reflecting MSA HHI. I report specifications including the baseline, time invariant, and modern controls, which leave the least amount of variation in the centralization measures available to be explained by the instrument.

Table 3.1: First Stage Estimates

	<i>Dependent variable:</i>		
	Top-Share	Number of Governments	MSA HHI
	(1)	(2)	(3)
RFD: Number of Closures	−3.550*** (1.090)	0.220*** (0.052)	−343.049*** (94.366)
Controls: Baseline	X	X	X
Controls: Time-Invariant	X	X	X
Controls: Modern	X	X	X
Observations	379	379	379
R ²	0.423	0.847	0.501
Adjusted R ²	0.377	0.835	0.461

*p<0.1; **p<0.05; ***p<0.01

Notes: First-stage regression estimates relating the Rural Free Delivery (RFD) post office closure instrument to three measures of local government centralization: Top-Share, the number of governments, and the MSA HHI. All specifications include baseline, time-invariant, and modern controls. Standard errors are reported in parentheses and the sample includes 379 MSAs.

Table 3.1 shows that post office closures induced by Rural Free Delivery between 1902 and 1910 strongly predict modern local government decentralization. The instrument decreases the percentage of employment in the largest government entity, increases the number of governments, and decreases the HHI of local government employment.

I report three main robustness checks. First, I examine whether the first-stage relationship differs systematically across metropolitan areas of different sizes. I split the sample at the median population and re-estimate the first-stage specifications separately for MSAs above and below the median population. The table is structured identically to Table 3.1, and only differs in the sample used. Panel A reports the first stage for a subsample containing the largest MSAs, and Panel B reports the first stage for a subsample containing the smallest MSAs.

Table 3.2: First Stage Estimates - Stratified by Population

	<i>Dependent variable:</i>		
	Top-Share	Number of Governments	MSA HHI
	(1)	(2)	(3)
<i>Panel A: Largest Metropolitan Areas</i>			
RFD: Number of Closures	−3.564** (1.654)	0.246*** (0.079)	−302.291** (125.515)
Controls: Baseline	X	X	X
Controls: Time-Invariant:	X	X	X
Controls: Modern	X	X	X
Observations	189	189	189
R ²	0.466	0.840	0.522
Adjusted R ²	0.372	0.812	0.439
<i>Panel B: Smallest Metropolitan Areas</i>			
RFD: Number of Closures	−3.254** (1.511)	0.190** (0.077)	−295.391** (142.232)
Controls: Baseline	X	X	X
Controls: Time-Invariant:	X	X	X
Controls: Modern	X	X	X
Observations	189	189	189
R ²	0.432	0.678	0.500
Adjusted R ²	0.333	0.621	0.413

*p<0.1; **p<0.05; ***p<0.01

Notes: First-stage regression estimates stratified by metropolitan population. Panel A reports results for the largest half of MSAs and Panel B reports results for the smallest half. The dependent variables are the three centralization measures: Top-Share, the number of governments, and the MSA HHI. All specifications include baseline, time-invariant, and modern controls.

Across both subsamples, the estimated effects of RFD-era closures on centralization closely resemble those obtained in the full sample, suggesting that the first stage effect is not being driven by population size heterogeneity.

I next examine geographic heterogeneity in the RFD instrument. In Table 3.3, I regress the centralization measures on interactions between the RFD instrument and Census Division fixed effects. Column 1 reports the results for the interaction on the Top-Share centralization measure with the baseline and time invariant controls and Column 2 adds the modern controls. Columns 3 and 4, along with Columns 5 and 6, repeat the same pattern for the Number of Governments and MSA HHI measures.

Table 3.3: First Stage Estimates - Geographic Heterogeneity

	<i>Dependent variable:</i>					
	Top-Share		Number of Governments		MSA HHI	
	(1)	(2)	(3)	(4)	(5)	(6)
RFD: Number of Closures x New England	−10.037*** (1.468)	−8.845*** (1.750)	0.609*** (0.072)	0.485*** (0.084)	−996.790*** (128.218)	−880.750*** (153.156)
RFD: Number of Closures x Middle Atlantic	−3.474*** (1.067)	−2.493** (1.243)	0.503*** (0.053)	0.377*** (0.060)	−419.162*** (93.171)	−329.880*** (108.738)
RFD: Number of Closures x South Atlantic	−4.576*** (0.899)	−3.744*** (1.066)	0.140*** (0.044)	0.102** (0.051)	−415.246*** (78.498)	−341.096*** (93.282)
RFD: Number of Closures x East South Central	−4.638*** (0.922)	−3.128*** (1.120)	0.254*** (0.045)	0.181*** (0.054)	−480.803*** (80.548)	−343.789*** (97.988)
RFD: Number of Closures x West South Central	−4.510*** (1.060)	−3.254*** (1.186)	0.271*** (0.052)	0.213*** (0.057)	−451.944*** (92.556)	−347.132*** (103.794)
RFD: Number of Closures x East North Central	−5.173*** (0.939)	−2.792** (1.210)	0.488*** (0.046)	0.315*** (0.058)	−507.973*** (81.968)	−297.954*** (105.849)
RFD: Number of Closures x West North Central	−5.465*** (1.090)	−2.753** (1.355)	0.453*** (0.054)	0.266*** (0.065)	−533.779*** (95.194)	−293.517** (118.608)
RFD: Number of Closures x Mountain	−5.809*** (1.748)	−2.967 (1.896)	0.237*** (0.086)	0.070 (0.091)	−516.682*** (152.642)	−261.435 (165.891)
RFD: Number of Closures x Pacific	−4.298*** (1.623)	−3.442** (1.742)	0.392*** (0.080)	0.280*** (0.084)	−435.127*** (141.728)	−321.688** (152.445)
Controls: Baseline	X	X	X	X	X	X
Controls: Time-Invariant:	X	X	X	X	X	X
Controls: Modern		X		X		X
Observations	379	379	379	379	379	379
R ²	0.340	0.407	0.813	0.840	0.420	0.477
Adjusted R ²	0.313	0.360	0.805	0.827	0.396	0.435

*p<0.1; **p<0.05; ***p<0.01

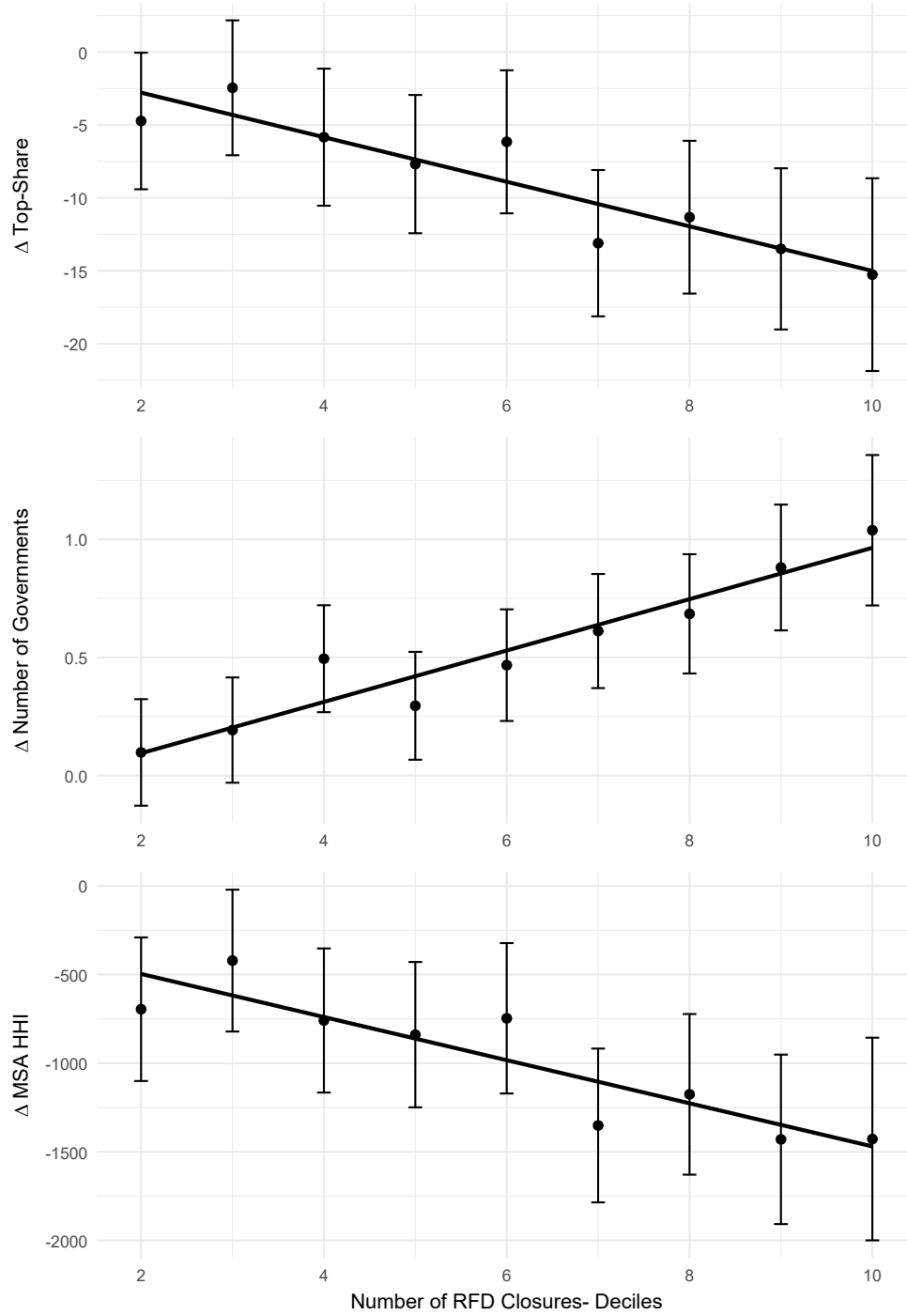
Notes: First-stage regression estimates allowing the effect of the Rural Free Delivery (RFD) instrument to vary across Census divisions. The dependent variables are the Top-Share measure, the number of governments, and the MSA HHI. Columns differ by control sets, with some specifications including modern controls in addition to baseline and time-invariant controls.

In the specifications with only the time invariant controls, the effect of the RFD instrument on centralization remains consistent with the results of Table 3.1 across all geographies. The effect of the instrument does vary across the United States, with the largest effects found in New England. In the specifications that include the modern controls, the results are largely still consistent, with exceptions in the Mountain division where the estimate loses precision. The table suggests that the variation induced by the instrument holds throughout the United States, and is not driven by geographic heterogeneity.

For a final robustness check on the instrument, I examine the effect of the first stage non-

parametrically. Figure 3.2 contains three panels, one for each centralization measure. To create each panel, the first-stage is re-estimated with the Baseline and Time Invariant controls, but the instrument is binned into deciles. The y-axis of the each panel reports the mean effect of the instrument on the centralization measure relative to the lowest decile with 95% confidence intervals. The x-axis reflects the ordering of the bins, where moving from left to right increases the number of post office closures induced by RFD in the bin.

Figure 3.2: Non-Parametric Effect of RFD Closures on Centralization



Notes: Non-parametric first-stage relationship between Rural Free Delivery (RFD) post office closures and measures of local government centralization. The instrument is grouped into deciles of closures. Each panel plots the estimated change in a centralization measure relative to the lowest decile with 95% confidence intervals. Panels correspond to Top-Share, the Number of Governments, and the MSA HHI.

The figure shows that a linear approximation captures the first-stage relationship well, suggesting that the linear functional-form restriction I impose in the IV specifications is unlikely to drive the results reported below. The figure also shows departures from perfect monotonicity. To describe these patterns, I use the language of Imbens and Angrist (1994), with the caveat that the figure does not identify individual level effects. The figure suggests that some MSAs appear to behave like “defiers”, in the sense that they experience many post office closures but remain relatively centralized, while others resemble “always-takers,” having relatively few closures yet still becoming decentralized.

Importantly, the MSAs that appear to behave like defiers or always-takers are not consistent across the three centralization measures. For example, the 6th decile in the Top-Share and MSA HHI measures shows a noticeable reversal in the effect of the instrument, while the Number of Governments measure remains comparatively well behaved in the same region of the instrument distribution. This heterogeneity across centralization measures, combined with their consistent agreement on the sign of the second-stage estimates reported below, suggests that the core findings do not hinge on a narrow portion of the instrument’s support.

4 Results

In this section, I present instrumental variables estimates of the effect of local government centralization. I begin by documenting the relationship between centralization and population growth. I then examine the effects of centralization on public employment and finances and assess whether these forces mediate the growth effects of centralization. I finally examine the effect of centralization on the location of data centers.

Outside of Tables 4.2, each table follows the same pattern. Columns 1 through 3 show the effects for the Top-Share centralization measure. Column 1 controls only for log population and log land area, Column 2 adds the time-invariant measures, and Column 3 adds the modern controls. Columns 4 through 6, along with Columns 7 through 9, follow the same pattern for the Number of Governments and MSA HHI measures of centralization, respectively.

Because the Top-Share measure is normalized to lie between 0 and 100, the coefficients presented indicate the effect of a 1 percentage point increase in the share of local government employment in the largest government entity. The Number of Governments measure is transformed by log base 2 meaning that coefficients reflect the estimated effect of doubling the number of governments in the MSA. The MSA HHI variable has been standardized, so coefficients express the effect of a one standard deviation increase in MSA HHI.

The tables report two complementary diagnostics commonly used in instrumental variables settings. First, the Kleibergen–Paap rk Wald F-statistic evaluates the relevance and strength of the excluded instrument in the first stage while remaining robust to heteroskedasticity. Larger values indicate stronger predictive power of the instrument for the endogenous regressor, and weak values suggest that conventional 2SLS inference may be unreliable (Kleibergen and Paap (2006)).

Second, the Anderson–Rubin test provides weak-instrument-robust inference for the estimated effect by testing whether a candidate value of the parameter is consistent with the

reduced-form relationship between the instrument and the outcome. The full procedure tests a set of candidate values by inverting a hypothesis test to create confidence sets. The p-value reported in the table corresponds to the p-value obtained when the candidate value is zero, with small values indicating that zero would not be included in the confidence set even without relying on a strong first stage (Anderson and Rubin (1949); Andrews, Stock, and Sun (2019)). Together, the KP statistic informs whether the instrument is sufficiently strong, while the AR test assesses whether the estimated effect remains statistically distinguishable from zero under inference procedures that remain valid even when instruments are weak.

4.1 MSA Level Population Growth

Table 4.1 reports two-stage least squares estimates of the effect of local government centralization on MSA population growth between 2000 and 2020.

Table 4.1: MSA Population Growth on Centralization - 2000 to 2020

	<i>Dependent variable:</i>								
	MSA Population Growth								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Top-Share	0.007*** (0.002)	0.004* (0.002)	0.006* (0.003)						
Number of Governments				-0.114*** (0.027)	-0.059* (0.034)	-0.106* (0.054)			
MSA HHI							0.095*** (0.024)	0.052* (0.030)	0.090** (0.045)
KP rk Wald F-statistic	45.75	26.14	11.36	40.98	34.21	16.06	54.28	30.94	14.07
Anderson–Rubin p-value	0	0.0679	0.0246	0	0.0679	0.0246	0	0.0679	0.0246
Controls: Baseline	X	X	X	X	X	X	X	X	
Controls: Time Invariant		X	X		X	X		X	X
Controls: Modern			X			X			X
Observations	379	379	379	379	379	379	379	379	379

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimates of the effect of local government centralization on metropolitan population growth between 2000 and 2020. Centralization is measured using Top-Share, the number of governments (log base 2), and the MSA HHI. Columns differ by control sets including baseline, time-invariant, and modern controls. Kleibergen–Paap rk Wald F-statistics and Anderson–Rubin p-values are reported for each specification. Standard errors are in parentheses.

The results displayed in the table are consistent with centralization increasing population growth. In the baseline specification, a one percentage point increase in Top-Share results in a .007 increase in log growth points. Doubling the number of governments, a move towards decentralization, decreases growth by .114 log growth points. Increasing MSA HHI

by a standard deviation increases the log population growth by .095 points. In the specification that adds the time-invariant controls, the effect decreases, but retains its sign. In the specification that adds the modern controls, the effects are largely in line with the baseline specification, but the variation used to identify the effects is less pronounced as indicated by the KP F-Statistics.

The decline in F-statistics when modern controls are added reflects how many modern variables are plausibly affected by local government centralization itself. An ideal example, that is not included as a modern control for this reason, is MSA level crime rates. All else equal, a potential resident would find higher expected utility in an MSA with a lower crime rate, meaning that crime rates are a likely determinant of population growth. But differences in crime rates across MSA are at least in part determined by the efficiency of the local policing system. If the efficiency of the local policing system is related to the organization of local government, variation in centralization induced by the instrument would be absorbed by crime rates, mechanically reducing the first stage relationship.

Many of the included modern controls have this feature to a lesser degree. For example, the percentage of households that rent is likely shaped by land use regulation that is in part determined by local government organization. For this reason, the specifications excluding modern controls are the preferred estimates for inference, while the fully conditioned models serve as a robustness check.

I next examine the effect of centralization on population growth throughout time. Table 4.2 contains three panels that differ only by the centralization measure used in the regression. Panel A focuses on the Top-Share centralization measure, Panel B on the Number of Governments measure, and Panel C on the MSA HHI measure. The columns in this table report growth rates measured using base years starting with 2000 and ending with 1950, with each decennial in between using the baseline and time invariant specifications. I annualize growth rates to make interpretation across years consistent.

Table 4.2: Centralization and Population Growth Through Time

	<i>Dependent variable:</i>					
	2000-2020	1990-2020	1980-2020	1970-2020	1960-2020	1950-2020
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Top-Share</i>						
Top-Share	0.0002* (0.0001)	0.0002* (0.0001)	0.0003** (0.0001)	0.0003** (0.0001)	0.0003* (0.0002)	0.0005** (0.0002)
KP rk Wald F-statistic	26.14	25.33	22.37	20.46	18.26	11.78
Anderson–Rubin p-value	0.0679	0.0607	0.0057	0.0181	0.0748	0.0075
Controls: Baseline	X	X	X	X	X	X
Controls: Time Invariant	X	X	X	X	X	X
Observations	379	379	379	379	379	379
<i>Panel B: Number of Governments</i>						
Number of Governments	−0.003* (0.002)	−0.003* (0.002)	−0.006** (0.002)	−0.006** (0.003)	−0.004 (0.003)	−0.010* (0.005)
KP rk Wald F-statistic	34.21	31.78	24.97	21.07	17.57	8.22
Anderson–Rubin p-value	0.0679	0.0607	0.0057	0.0181	0.0748	0.0075
Controls: Baseline	X	X	X	X	X	X
Controls: Time Invariant	X	X	X	X	X	X
Observations	379	379	379	379	379	379
<i>Panel C: MSA HHI</i>						
MSA HHI	0.003* (0.001)	0.003* (0.002)	0.005** (0.002)	0.004** (0.002)	0.004* (0.002)	0.007** (0.003)
KP rk Wald F-statistic	30.94	30.21	26.93	24.82	22.4	14.13
Anderson–Rubin p-value	0.0679	0.0607	0.0057	0.0181	0.0748	0.0075
Controls: Baseline	X	X	X	X	X	X
Controls: Time Invariant	X	X	X	X	X	X
Observations	379	379	379	379	379	379

*p<0.1; **p<0.05; ***p<0.01

Notes: Each column reports a separate instrumental variables regression of MSA population growth between the base year listed in the column header and 2020. Panel A uses the employment share of the largest local government (Top-Share), Panel B uses the number of governments in the MSA, and Panel C uses the Herfindahl–Hirschman Index (HHI) of local government employment as measures of centralization. All specifications include baseline and time-invariant controls. Kleibergen–Paap rk Wald F-statistics and Anderson–Rubin p-values are reported to evaluate instrument strength and weak-instrument robust inference. Standard errors are reported in parentheses.

The results of the table suggest that the positive effect of centralization is relatively consistent throughout time. However, the validity of these results rely on the assumption that the organization of local government measured in 2007 accurately measures the organization in the past. This assumption is likely valid for more modern base years, but may not hold for periods further back in history. The monotonic decrease in the F-Statistics throughout all three measures is suggestive of this pattern.

Interpreted through a spatial equilibrium framework, the results of Tables 4.1 and 4.2 imply that centralization increases residents expected utility within MSAs, attracting population over time. This finding contrasts with decentralization-based theories that suggest competition between jurisdictions will provide market-like discipline in the supply of public goods, leading to higher population growth. In the remaining empirical exercises, I look to examine the mechanisms that promote the higher growth found in centralized MSAs.

4.2 Local Government Employment

I now examine how local government centralization affects local government employment. Table 4.3 reports the instrumental variables estimates for the effect of centralization on log local government employment. The table is organized similarly to Table 4.1.

Table 4.3: Local Government Employment on Centralization

	<i>Dependent variable:</i>								
	Log Local Government Employment								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Top-Share	0.009*** (0.003)	0.012*** (0.004)	0.001 (0.005)						
Number of Governments				-0.137*** (0.042)	-0.188*** (0.060)	-0.022 (0.073)			
MSA HHI							0.120*** (0.035)	0.166*** (0.055)	0.019 (0.063)
KP rk Wald F-statistic	48.42	26.83	11.74	46.59	35.6	17.85	57.45	31.69	14.48
Anderson–Rubin p-value	4e-04	5e-04	0.7652	4e-04	5e-04	0.7652	4e-04	5e-04	0.7652
Controls: Baseline	X	X	X	X	X	X	X	X	
Controls: Time Invariant		X	X		X	X		X	X
Controls: Modern			X			X			X
Observations	379	379	379	379	379	379	379	379	379

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimates of the effect of local government centralization on log local government employment across metropolitan statistical areas (MSAs). Centralization is measured using Top-Share, the number of governments (log base 2), and the MSA HHI. Columns differ by control sets including baseline controls, time-invariant controls, and modern controls. Kleibergen–Paap rk Wald F-statistics and Anderson–Rubin p-values are reported for each specification. Standard errors are reported in parentheses.

Table 4.3 shows that centralization of local government causes higher local government employment. A one percent increase in Top-Share results in a .9 or 1.2% increase in employment depending on the specification. Doubling the number of governments decreases local government employment by 13 or 18%. A one standard increase in MSA HHI results in local government employment increasing by either 12 or 16%.

These findings begin to narrow the set of plausible mechanisms underlying the gains from centralization. If economies of scale driven by technical efficiency were the primary channel, centralization would be expected to reduce labor inputs. The results instead show higher employment under greater centralization, which is inconsistent with that mechanism.

I next examine total MSA level tax revenues. Table 4.4 mirrors Table 4.3, but with log local government tax revenue as the dependent variable and median income included in all specifications to account for differences in local price levels across MSA.

Table 4.4: Local Government Tax Revenue on Centralization

	<i>Dependent variable:</i>								
	Log Local Government Tax Revenue								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Top-Share	0.012*** (0.004)	0.020*** (0.006)	0.013 (0.008)						
Number of Governments				-0.182*** (0.056)	-0.313*** (0.082)	-0.204* (0.115)			
MSA HHI							0.165*** (0.055)	0.277*** (0.080)	0.176 (0.108)
KP rk Wald F-statistic	50.43	26.79	11.74	47.58	34.9	17.85	58.66	31.74	14.48
Anderson–Rubin p-value	0.0013	0	0.0698	0.0013	0	0.0698	0.0013	0	0.0698
Controls: Baseline	X	X	X	X	X	X	X	X	
Controls: Time Invariant		X	X		X	X		X	X
Controls: Modern			X			X			X
Observations	379	379	379	379	379	379	379	379	379

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimates of the effect of local government centralization on log local government tax revenue across MSAs. Centralization is measured using Top-Share, the number of governments (log base 2), and the MSA HHI. Median income is included in all specifications to account for differences in the value of a dollar across metropolitan areas. Columns differ by the inclusion of baseline, time-invariant, and modern control variables. Kleibergen–Paap rk Wald F-statistics and Anderson–Rubin p-values are reported for each specification. Standard errors are reported in parentheses.

Table 4.4 shows that more centralized MSAs collect more tax revenue. A one percentage point increase in Top-Share causes an increase in tax revenues around 1.2%. Doubling the number of local governments in an MSA reduces tax revenues by roughly 18%, with the opposite effect for a standard deviation increase in MSA HHI.

These results are largely consistent with the predictions of tax competition models which represent an important coordination failure. In these models, decentralization increases the amount of competition between local governments. When setting tax rates, at the margin, individual local governments focus on how increases in their own tax rate decrease their tax base, but do not internalize that the lost tax base is largely shifted to other jurisdictions within the MSA. As a result, equilibrium tax rates and total local government tax revenue

is predicted to be lower in decentralized MSAs than in centralized MSAs where competition is less intense.

To further distinguish between the mechanisms of economies of scale and coordination failures, I examine if the effect of centralization on population growth is mediated by local government employment or tax revenue. Because centralization is instrumented, I can separately identify the effects of centralization from the effect on either outcome. The effects of the two potential mediators are reported in Table 4.5 which recreates Table 4.1 with the addition of the potential mediator in all specifications.

Table 4.5: Mechanisms for Population Growth

	<i>Dependent variable:</i>								
	MSA Population Growth: 2000 to 2020								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Local Government Employment									
Top-Share	0.007*** (0.002)	0.004* (0.002)	0.006* (0.003)						
Number of Governments				−0.113*** (0.031)	−0.061* (0.035)	−0.109* (0.064)			
MSA HHI							0.093*** (0.028)	0.055* (0.032)	0.082* (0.047)
Local Government Employment	0.006 (0.045)	−0.017 (0.036)	0.053 (0.037)	0.001 (0.042)	−0.008 (0.034)	0.071* (0.039)	0.009 (0.043)	−0.016 (0.036)	0.053 (0.037)
KP rk Wald F-statistic	35.16	22.79	10.17	29.46	33.07	10.92	42.72	27.78	12.81
Anderson–Rubin p-value	3e-04	0.0711	0.0497	3e-04	0.0711	0.0497	3e-04	0.0711	0.0497
Observations	379	379	379	379	379	379	379	379	379
Panel B: Local Government Tax Revenue									
Top-Share	0.005*** (0.002)	0.003 (0.002)	0.006* (0.003)						
Number of Governments				−0.089*** (0.030)	−0.053 (0.038)	−0.113 (0.071)			
MSA HHI							0.070*** (0.024)	0.043 (0.031)	0.078* (0.046)
Tax Revenue	0.086*** (0.022)	0.030 (0.023)	0.057** (0.023)	0.034 (0.029)	0.018 (0.027)	0.021 (0.033)	0.090*** (0.021)	0.027 (0.024)	0.049** (0.023)
KP rk Wald F-statistic	49.23	24.07	10.4	29.53	25.41	9.18	54.32	27.12	12.45
Anderson–Rubin p-value	0.0036	0.1478	0.0651	0.0036	0.1478	0.0651	0.0024	0.1478	0.0651
Observations	379	379	379	379	379	379	379	379	379

*p<0.1; **p<0.05; ***p<0.01

Notes: Instrumental variables regressions examining whether local government employment and tax revenue account for the relationship between centralization and population growth. Panel A includes log local government employment as an additional regressor, while Panel B includes log local government tax revenue. Centralization is measured using Top-Share, the number of governments (log base 2), and the MSA HHI. Columns differ by the inclusion of baseline, time-invariant, and modern controls. Kleibergen–Paap rk Wald F-statistics and Anderson–Rubin p-values are reported for instrument diagnostics. Standard errors are reported in parentheses.

In all but one specification in Table 4.5 Panel A, log local government employment has no statistical relationship with growth and enters with differing signs across specifications. Further, the effect of centralization on growth is largely consistent with the results found in

Table 4.1 where local government employment is not included. Because I do not directly observe public good output, I cannot fully test economies of scale as a broader mechanism. However, the evidence narrows the set of plausible scale channels, particularly to those that do not operate through changes in local government employment levels.

In contrast, Table 4.5 Panel B shows that tax revenue has a positive and often statistically significant association with population growth. The estimated effects of centralization decline modestly and lose precision in some specifications with the inclusion of revenue. I fail to rule out that coordination failures, including tax competition, are a mechanism behind the higher population growth found in more centralized MSAs. One possible interpretation is that centralization increases fiscal capacity, and fiscal capacity accounts for the increased growth found in centralized MSAs.

As I do not have a separate instrument for local government employment or tax revenue, the estimates for the effect of those variables on growth presented in Table 4.5 are only suggestive of the dominant mechanism, not causal.

4.3 Data Center Locations

In this section, I examine the effect of local government centralization on the location decisions made by firms when building data centers. Table 4.6 presents the IV estimates following the pattern of the previous tables.

Table 4.6: Data Center Location on Centralization

	<i>Dependent variable:</i>								
	Data Center in MSA								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Top-Share	−0.007* (0.004)	−0.011* (0.006)	−0.011 (0.010)						
Number of Governments				0.107* (0.060)	0.177* (0.095)	0.171 (0.146)			
MSA HHI							−0.099* (0.057)	−0.161* (0.088)	−0.153 (0.134)
KP rk Wald F-statistic	51.08	27.21	12.16	53.87	37.71	20.31	61.08	32.27	15.01
Anderson–Rubin p-value	0.0671	0.0474	0.2198	0.0671	0.0474	0.2198	0.0671	0.0474	0.2198
Controls: Baseline	X	X	X	X	X	X	X	X	
Controls: Time Invariant		X	X		X	X		X	X
Controls: Modern			X			X			X
Observations	379	379	379	379	379	379	379	379	379

*p<0.1; **p<0.05; ***p<0.01

Notes: Instrumental variables estimates of the effect of local government centralization on the probability that a metropolitan statistical area hosts at least one data center facility. Centralization is measured using Top-Share, the number of governments (log base 2), and the MSA HHI. Columns differ by the inclusion of baseline, time-invariant, and modern control variables. Kleibergen–Paap rk Wald F-statistics and Anderson–Rubin p-values are reported for instrument diagnostics. Standard errors are reported in parentheses.

Across the baseline and time-invariant specifications, higher centralization is associated with a lower likelihood of data-center presence. A one percentage point increase in Top-Share reduces the probability that an MSA hosts a data center by approximately 0.7 to 1.1 percentage points. Similarly, a one standard deviation increase in MSA HHI lowers the probability of hosting a data center by roughly 10 to 17 percentage points. In contrast, doubling the number of governments raises the probability by approximately the same amount. Although the estimates become less precise in the most saturated specifications, the direction and scale of the effects remains consistent across centralization measures, suggesting that more fragmented governance structures are associated with a higher probability of attracting data-center investment.

The results in Table 4.6 should be considered alongside the results of Table 4.4. Table 4.4 suggests that decentralization intensifies tax competition, resulting in lower tax rates and less tax revenue taken in by local governments. Tax sensitive firms then have an incentive to act on this information and choose to locate in more fragmented MSAs to take advantage of the lower effective tax burdens. The estimates show that decentralized MSAs are significantly more likely to host data centers, a form of highly mobile capital that is particularly sensitive to local fiscal conditions. The finding that decentralized MSAs host more data centers is then consistent with the models of tax competition and provides further causal evidence that coordination failures like tax competition are likely to be a dominant mechanism behind the

population gains from centralization.

5 Conclusion

This paper estimates the causal effect of local government centralization on metropolitan population growth. Using variation across U.S. metropolitan statistical areas and an instrumental variables strategy based on post office closures induced by the introduction of Rural Free Delivery, I show that greater local government centralization increases population growth, challenging theories of local government that advocate for decentralization. Centralization also leads to higher levels of local government employment and tax revenue. Local government employment is not found to affect population growth when centralization is accounted for, but tax revenue does, which suggests that the increase in population growth is more likely to be driven by coordination failures than economies of scale. Finally, decentralized MSAs are found to be more likely to host data centers. Decisions on where to site data centers are particularly sensitive to fiscal factors, which under standard tax competition models are more favorable to firms in decentralized MSAs, suggesting that coordination failures are salient enough to affect firm decision-making.

The evidence regarding the mechanisms driving population growth in centralized MSAs points most strongly to the resolution of coordination failures. Local governments operate in close spatial proximity within metropolitan areas and disproportionately supply goods and services characterized by externalities. Under decentralized governance, these externalities generate coordination failures. By internalizing these externalities, centralized governance improves coordination, which increases the efficiency of the supply of public goods, increasing residents' expected utility, and accounting for the higher population growth found in centralized MSAs.

These results also suggest several avenues for future research and potential applications. More broadly, they speak to the design of local government institutions within metropolitan areas. If the primary gains from centralization arise from improved coordination rather than cost-saving efficiencies, then policies that increase cohesion in tax bases, regulatory authority, or service provision across jurisdictions may yield benefits even in the absence of formal consolidation.

The results also shed light on competition across different levels of government. The bidding-for-firms literature (Black and Hoyt (1989); Bartik (1991); Glaeser (2001); Garcia-Milà and McGuire (2002); Greenstone and Moretti (2003); Kim (2025); Ferrari and Ossa (2023); Slatery (2025)) focuses on competition between MSAs for mobile firms. This paper contributes two points to this literature. First, the organization of local government seems to have important effects. Models that treat aggregates like MSAs as single decision makers may be masking important heterogeneity between them. Second, this literature emphasizes both the potential efficiency gains and welfare losses associated with subsidy competition. One interpretation of the findings here is that, within MSAs, the welfare costs of coordination failures outweigh the benefits of competition among local governments. At broader geographic scales, however, the relevant tradeoffs may differ in the sense that externalities are less prominent,

and competition across metropolitan areas may be welfare enhancing. Determining if, and if so where, these margins flip remains an important topic for future work.

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